Probability with Counting KEY

Name _____

When every outcome is equally likely, count the outcomes with permutations or combinations.

$$P(\text{event}) = \frac{\text{number of ways the event can happen}}{\text{total number of possible outcomes}}$$ Count both with the same method.

Example 1: choosing a group. A teacher randomly picks 2 students from 5 girls and 4 boys. What is P(both girls)?

Total ways to pick 2 of 9: ${}_9C_2 = 36$

Ways to pick 2 of the 5 girls: ${}_5C_2 = 10$

Probability: $P(\text{both girls}) = \frac{10}{36} = \frac{5}{18} \approx 0.28$

Example 2: a lottery. Pick 3 different numbers from 1 to 20. Order does not matter. What is P(win)?

Total: ${}_{20}C_3 = 1{,}140$. Only 1 of them wins. $P(\text{win}) = \frac{1}{1{,}140} \approx 0.0009$

Example 3: jobs. 10 students are in a club. A president and a vice president are chosen at random. What is the probability that Ana is president and Ben is vice president?

Order matters (different jobs). Total: ${}_{10}P_2 = 90$. Only 1 way works. $P = \frac{1}{90}$

Steps

1. Decide: does order matter? (permutation or combination)
2. Count all the possible outcomes.
3. Count the outcomes you want.
4. Divide.

Work with a partner. Compare answers when you finish each problem.

1. A committee of 3 is chosen at random from 6 juniors and 4 seniors.

   (a) How many committees are possible? ${}_{10}C_3 = 120$

   (b) Find P(all seniors). ${}_4C_3 = 4$. $\frac{4}{120} = \frac{1}{30}$

   (c) Find P(all juniors). ${}_6C_3 = 20$. $\frac{20}{120} = \frac{1}{6}$

2. A lottery: pick 4 different numbers from 1 to 30. Order does not matter. What is the probability of winning with one ticket?

   ${}_{30}C_4 = 27{,}405$. $P(\text{win}) = \frac{1}{27{,}405}$

3. 6 runners are in a race. Each order of finish is equally likely. What is the probability that Maya finishes 1st and Leo finishes 2nd?

   Order matters. ${}_6P_2 = 30$. $P = \frac{1}{30}$

4. Two cards are dealt from a deck of 52. What is P(both aces)? (There are 4 aces.)

   ${}_4C_2 = 6$ and ${}_{52}C_2 = 1{,}326$. $P = \frac{6}{1{,}326} = \frac{1}{221}$

5. In a taste test, a person tastes 4 drinks and guesses which brand each one is. Each brand is used once. If the person just guesses, what is the probability that all 4 are right?

   There are $4! = 24$ ways to match the brands. Only 1 is right. $P = \frac{1}{24}$

6. You grab 2 shirts at random from a drawer of 8 shirts. 3 of the shirts are blue. What is P(both blue)?

   ${}_3C_2 = 3$ and ${}_8C_2 = 28$. $P = \frac{3}{28}$

7. A big lottery asks you to pick 5 numbers from 1 to 70 and 1 extra number from 1 to 25.

   (a) Use a calculator. How many different tickets are possible?

   ${}_{70}C_5 \cdot 25 = 12{,}103{,}014 \cdot 25 = 302{,}575{,}350$

   (b) What is the probability of winning with one ticket? Is it a good idea to expect to win?

   $\frac{1}{302{,}575{,}350}$. No. Winning is extremely unlikely.